
CS 542 : Design Patterns for Scalable Systems

Course Code: CS 542

Course Name : Design Patterns for Scalable Systems

L-T-P-C : 3-0-0-3

Intended for : B. Tech./Masters/Ph.D.

Prerequisite : CS 310 and CS 309 or equivalent courses covering basics of computer networks, databases and operating system

Mutual Exclusion : None

Approval: 45th BoA

Course Contents:

- **Introduction to design patterns and reactive systems:** Software design: functional and non-functional requirements, design patterns, changing landscape: web-scale responsive applications, design approach: monolithic vs microservices, horizontal vs vertical scaling, walk-through of the reactive manifesto. (5 hours)
- **Review of concurrency in practice:** Threads in OS, notions of parallelism: tasks, functions, and loops, synchronization and data flow concepts, consistency protocols, transition from parallel to distributed systems: distributed memory architectures, CAP theorem, distributed state: sharding and conflict free replicated data types. (6 hours)
- **Review of functional programming concepts:** Immutability, pattern matching, partial functions, collections, futures (5 hours)
- **Actor model:** Asynchronous programming, Actor model, message processing semantics, designing and testing Actor models (7 hours)
- **Design patterns for high availability:** Fault-tolerance and recovery patterns, replication patterns, resource management patterns, state management and persistence patterns (14 hours)
- **Design patterns for effective communication:** Message flow patterns, flow control patterns (5 hours)

Laboratory/practical/tutorial Modules:

None

Text books:

1. Roland Kuhn, Brian Hanafey and Jamie Allen, **Reactive Design Patterns**, Manning Publications Co., 2017.
2. Erich Gamma, Richard Helm, Ralph Johnson and John Vlissides, **Design Patterns: Elements of Reusable Object-Oriented Software**, Addison-Wesley, 1995.



References:

1. Clement Escoffier and Ken Finnigan, **Reactive Systems in Java**, 1st Edition, O'Reilly, 2021.
2. Martin Kleppmann, **Designing Data-Intensive Applications**, 17th Release, O'Reilly, 2021.
3. Chris Richardson, **Microservices Patterns with examples in Java**, Manning Publications Co., 2018.
4. Jonas Boner, **Reactive Microsystems: The Evolution of Microservices at Scale**, 1st Edition, O'Reilly, 2017.
5. Thomas Anderson and Dahlin, **Operating Systems: Principles and Practice**, 2nd Edition, Recursive books, 2011.